

Project 8

RecklamRadar – Cross-Platform Deal & Store Management Application

Flutter + Firebase Multi-Platform Application

1. Project Overview

RecklamRadar is a cross-platform mobile and web application developed using Flutter. The system functions as a deal discovery and store management platform where users can browse store deals, manage accounts, and interact with listings, while administrators can manage store content and product data.

The application was designed to run across:

- Android
- iOS
- Web
- macOS
- Linux

This demonstrates multi-platform engineering capability using a single codebase.

2. Core Objective

The goal of RecklamRadar was to build a production-ready, cloud-connected application that includes:

- User authentication
- Role-based access control (User / Admin)
- Store listing management
- Product/deal addition
- Account management

- Firebase backend integration
- Responsive UI design

The system combines frontend UI logic with cloud database connectivity and real-time backend services.

3. System Architecture

The application follows a client-cloud architecture:

Frontend:

- Flutter (Dart)
- Stateful and stateless widget structure
- Provider for theme/state management

Backend:

- Firebase Authentication
- Firestore database
- Cloud configuration files
- Data migration scripts

The architecture separates:

- Presentation layer (UI widgets)
 - Business logic
 - Firebase service layer
 - Role-based navigation control
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4. Authentication & Role Management

Authentication is handled using Firebase Authentication.

Workflow:

1. User registers or logs in.
2. Firebase verifies credentials.
3. Role is determined (admin or regular user).
4. Navigation is conditionally controlled.

Admin users can:

- Access admin dashboard
- Manage store data
- Add or modify listings

Regular users can:

- Browse deals
- View store pages
- Manage account details

This demonstrates secure role-based UI separation.

5. Core Features

The application includes:

- Login & registration screens
- Password recovery functionality
- Home dashboard
- Store deals page
- Cart interface
- Item adding page (admin)
- Account details page
- Settings page
- Theme switching

State management ensures consistent UI updates.

6. Database Design (Firestore)

Firestore collections likely include:

- Users
- Stores
- Deals
- Products
- Cart items

Data is structured to allow:

- Store-based filtering
- Deal categorization
- User-specific data retrieval
- Admin data management

Firebase ensures:

- Cloud-hosted backend
 - Scalable data storage
 - Real-time updates capability
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7. Multi-Platform Engineering

The project includes configuration for:

- Android (Gradle setup)
- iOS (Xcode configuration)
- Web (Flutter web build)
- macOS
- Linux

This shows:

- Platform adaptation
 - Environment configuration knowledge
 - Cross-platform UI responsiveness
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8. Deployment & Configuration

The project includes:

- Firebase configuration files
- google-services.json
- Firebase options configuration
- Data migration scripts

This indicates readiness for:

- Production deployment
- Cloud hosting
- Remote data management

9. Technical Skills Demonstrated

- Flutter cross-platform development
- Firebase integration
- Role-based access control
- Cloud-connected application architecture
- Stateful widget management
- Responsive UI design
- Environment configuration
- Mobile + Web build systems

10. Real-World Applicability

This architecture can scale into:

- Retail deal aggregation apps
- Marketplace mobile platforms
- Store management dashboards
- Startup MVP applications
- Cloud-based retail applications



All Projects Completed

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1. Thesis (Deep Learning + CV)
 2. Adaptive AI Security Research
 3. SQL vs ORM Performance Study
 4. SkillSwap (Full Stack)
 5. ScandiCart (Full Stack + Agile)
 6. Lead Ingestion Automation
 7. Knowledge Exchange SRD
 8. RecklamRadar (Flutter)
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